



Grocery Basket Analysis

Instacart Grocery Basket

Project overview

Instacart is an online grocery store using an app. Although Instacart currently has very strong sales, they still wish to learn more about their sales trends. They're thinking about a targeted marketing approach because they believe they can't target everyone with the same techniques.

Key questions

- What days of the week and hours of the day are busiest?
- What times of the day do people spend the most money?
- Are there certain types of products that are more popular than others?
- What's the distribution among users in regard to their brand loyalty?
- Are there differences in ordering habits based on a customer's loyalty status?
- Are there differences in ordering habits based on a customer's region?
- Is there a connection between age and family status in terms of ordering habits?
- What differences can you find in ordering habits of different customer profiles? consider the price of orders, frequency of orders, products customers are ordering, and anything else you can think of.

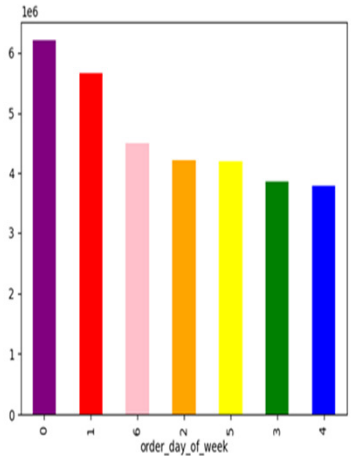
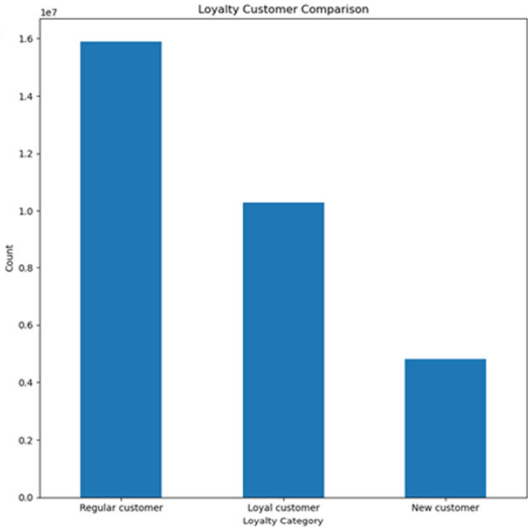
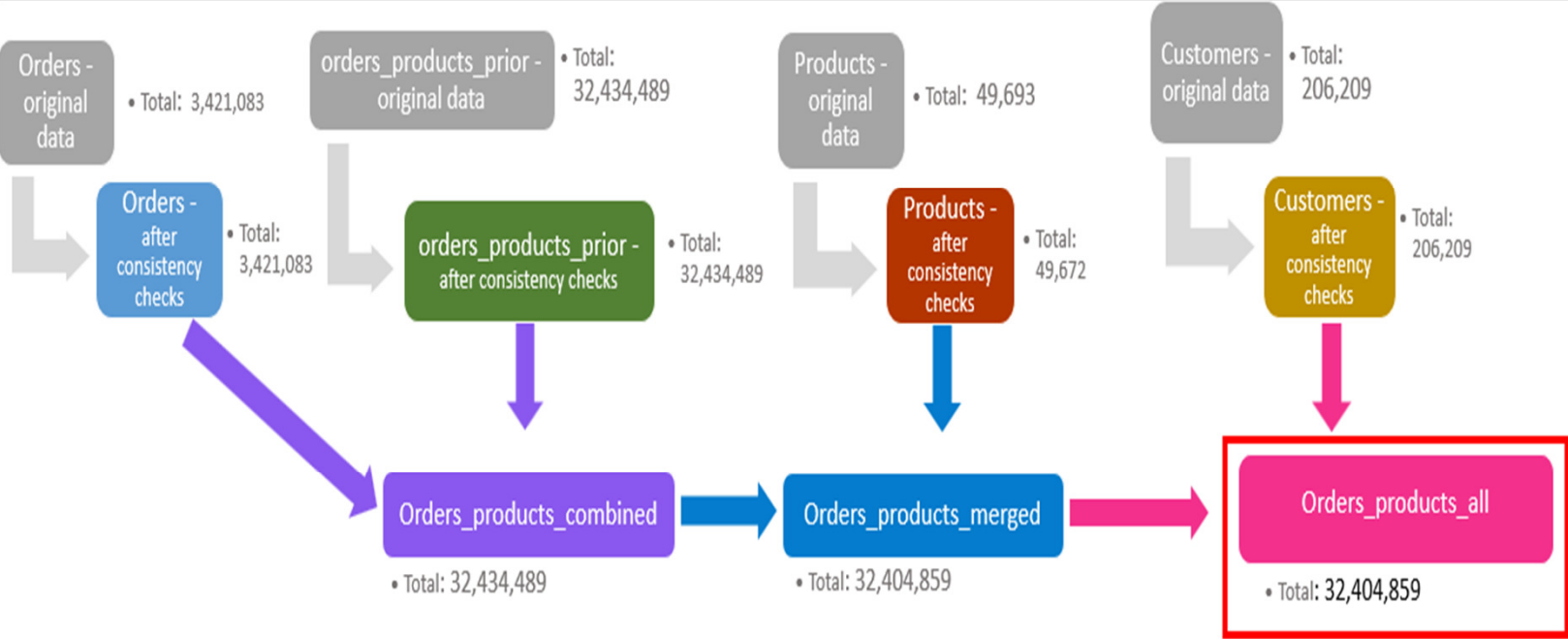
Aims

- Conduct a preliminary data and exploratory analysis to identify trends in Instacart's sales in order to draw conclusions and make recommendations for methods for better customer segmentation.

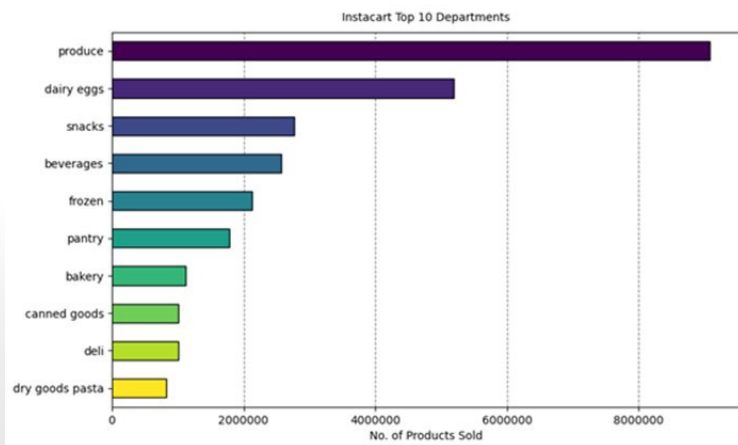
Skills

- Installed required Python libraries like Pandas, NumPy, Matplotlib, and Seaborn
- Grouping, aggregating Data and Visualization in Python

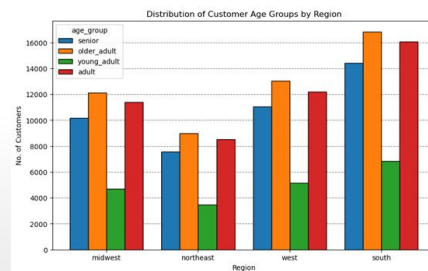
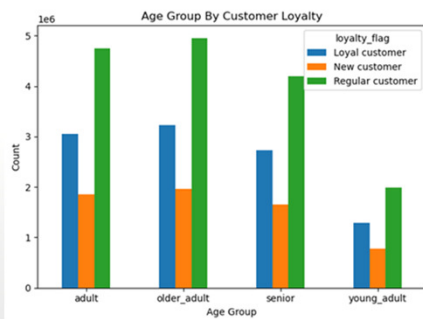
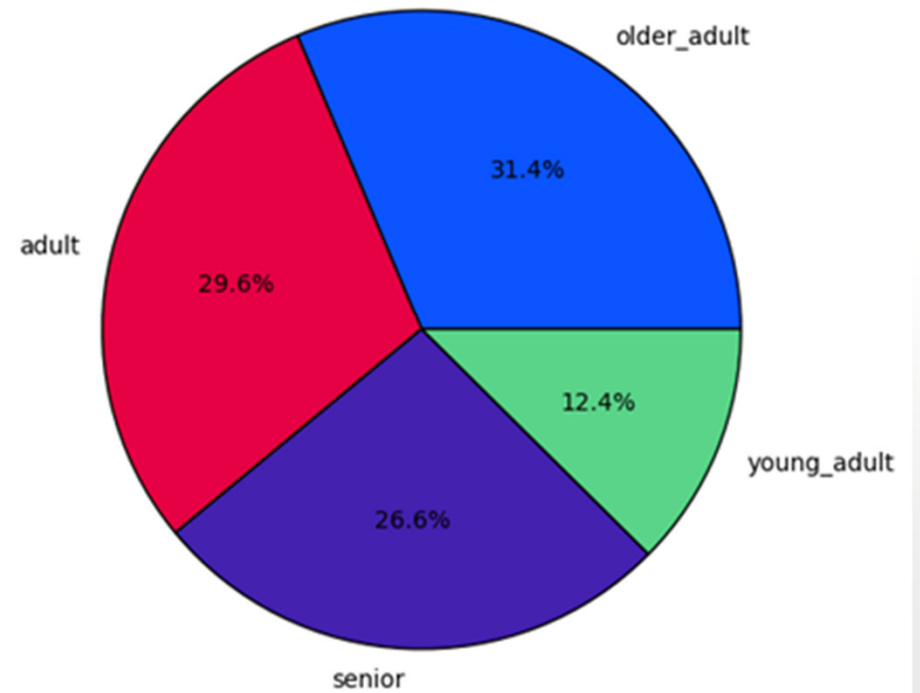
Population flow



- 1.) The grey boxes in the first row of the population flow represent the original data sets as they were when you downloaded them. In the Total fields you need to add the count of the rows when you imported the data set into Jupyter.
- 2.) The second row of boxes (coloured) represents the data sets **after** you manipulated them, e.g., removed missing values and duplicates. In the Total fields you need to add the count of the rows after conducting these operations. This offers a visual overview of how the data *flows* throughout the data consistency checks.
- 3.) The third row, where also the arrows are coloured, represents the merges you performed between the datasets. In the Total fields you need to add the count of the rows in the merged datasets, so that you end up with the final dataset (in the red box). Keep in mind the final dataset should be without exclusions (based on the exclusion flag).



Instacart_project_distribution of Customers by Age Category



Insight

Business activity

- The highest number of orders are placed between 10 a.m. and 3 p.m. on Saturday and Sunday.
- These are the busiest days of the week.
- Wednesday morning is the least busy day of the week.
- Between 8 a.m to 4 p.m is when customers spend most though, 9 a.m is the peak hour.

Product sales

- Produce, dairy and eggs, snacks, drinks, and frozen food are the top five products.

Customers analysis

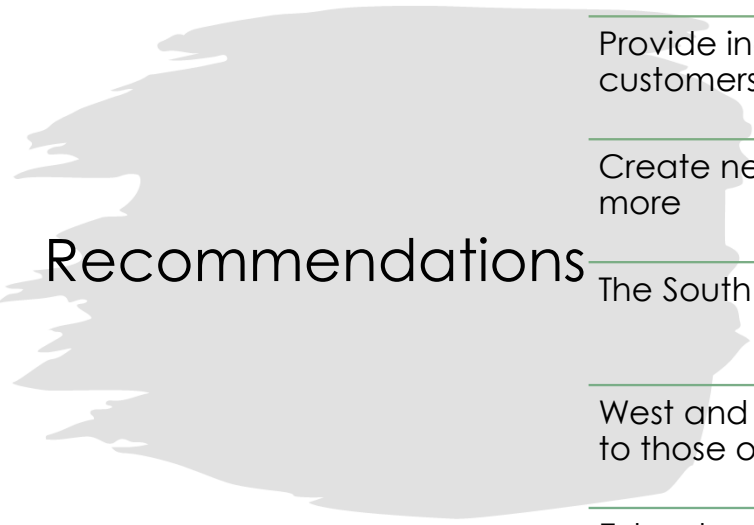
- Regular customers outnumber loyal customers in this case.

Spending habit

- Low and high spenders make up the majority of all loyalty status groups.
- Although there seem to be more high-spenders among regular customers, their percentage is small in comparison to the other groups.

Demography and income status of customers

- In every age group, the South has the greatest number of customers, followed by the West, the Midwest, and the Northeast.
- There is no particular relationship found between age and family status.
- The majority of Instacart's regular, non-regular, and frequent customers are married.
- Singles, divorced or widowed people, and people who live with their parents and siblings are the next most common demographics.
- Customers with incomes up to 200,000 are concentrated among all age groups, and those 40 years of age and above have incomes up to 600,000.
- Customers 40 years of age and older may spend more.



Recommendations

Run their advertising campaigns during off-peak times, from 12 to 5 AM, and schedule extra ads for Tuesdays and Wednesdays.

Provide exclusive deals on those categories of produce like dairy and eggs, snacks, drinks, and frozen food to maintain customer interest and engagement.

Provide incentives to its loyal and new customers to encourage them to become regular customers as well as to reward its regular customers.

Create new incentives like bonus points or amount-based discounts for customers to spend more

The South should be the primary target for Instacart's advertising.

West and Midwest should receive more attention since their ordering practices are similar to those of the South.

Extend exclusive promotions to senior citizens to encourage them to place more orders.

Offering products that young adults are interested in at special prices or discounts will also encourage them to spend more money.